General Questions

1. **Q: What is the goal of this project?**

   - **A:** The goal of this project is to build a machine learning model to detect fraudulent credit card transactions. It aims to minimize false positives and false negatives in identifying fraudulent activities in financial transactions.

2. **Q: What dataset did you use for this project?**

   - **A:** The project typically uses the publicly available **Credit Card Fraud Detection dataset** from Kaggle, which contains transactions made by European cardholders in September 2013, with 284,807 transactions.

3. **Q: What are the features in the dataset?**

   - **A:** The dataset contains 30 features: 28 anonymized features (V1–V28), `Time`, and `Amount`. Additionally, there is a `Class` column, where '1' represents fraud and '0' represents non-fraud.

4. **Q: Why are the features anonymized in the dataset?**

   - **A:** The features are anonymized to protect sensitive customer information and maintain privacy as per data protection regulations.

5. **Q: How is the dataset imbalanced?**

   - **A:** The dataset is highly imbalanced, with only about 0.172% of the transactions labeled as fraudulent.

Technical Questions

6. **Q: How did you handle the class imbalance in the dataset?**

   - **A:** Techniques such as **undersampling** the majority class, **oversampling** the minority class (e.g., SMOTE), or using algorithms that are robust to class imbalance (e.g., Random Forest, XGBoost) were used.

7. **Q: Which machine learning models did you use?**

- **A:** I experimented with models such as Logistic Regression, Random Forest, Gradient Boosting, and Neural Networks. Random Forest and Gradient Boosting often performed better due to their robustness against imbalanced data.

8. **Q: What preprocessing steps did you perform?**

- **A:** Preprocessing included:

  - Scaling the `Amount` and `Time` features using techniques like **StandardScaler** or **MinMaxScaler**.

  - Splitting the dataset into training and testing sets.

  - Handling class imbalance using resampling techniques.

9. **Q: What evaluation metrics did you use?**

- **A:** Metrics like **Precision**, **Recall**, **F1-Score**, and **ROC-AUC** were used because of the imbalanced nature of the dataset. Accuracy was not prioritized as it can be misleading for imbalanced data.

10. **Q: What was your final model's performance?**

- **A:** The final model achieved a high **ROC-AUC score** and a balance between Precision and Recall, ensuring that fraudulent transactions were detected with minimal false positives and false negatives.

Domain-Specific Questions

11. **Q: Why is fraud detection important in credit card transactions?**

- **A:** Fraud detection is critical to prevent financial losses for both customers and financial institutions and to maintain trust in digital payment systems.

12. **Q: What are the challenges in fraud detection?**

- **A:** Key challenges include:

  - The highly imbalanced nature of the data.

  - Evolving patterns in fraudulent activities.

  - The need for real-time detection to prevent losses.

13. **Q: Why did you choose machine learning for this project?**

- **A:** Machine learning can analyze large datasets efficiently, learn complex patterns, and adapt to evolving fraud patterns, making it ideal for fraud detection.

Advanced Questions

14. **Q: How would you deploy this model in a production environment?**

- **A:** The model can be deployed using tools like Flask or FastAPI to create an API endpoint. It can then be integrated with the transaction processing system to flag suspicious transactions in real-time.

15. **Q: How can you improve the model further?**

- **A:** Improvements could include:

  - Using ensemble methods like stacking or boosting.

  - Incorporating temporal features or domain-specific features.

  - Leveraging deep learning techniques for feature extraction.

16. **Q: How do you ensure the model generalizes well?**

- **A:** Techniques like cross-validation, proper train-test splits, and testing on unseen data help ensure the model generalizes well.

17. **Q: How would you handle false positives in fraud detection?**

- **A:** False positives can be reduced by tuning the model's threshold or using a two-stage detection system where flagged transactions undergo additional verification.

Behavioral Questions

18. **Q: What was the most challenging part of this project?**

   - **A:** Handling the severe class imbalance was challenging. I had to experiment with various resampling techniques and evaluation metrics to optimize the model's performance.

19. **Q: How did you ensure the project’s reproducibility?**

   - **A:** I ensured reproducibility by version-controlling the code using GitHub, documenting the steps in the Jupyter Notebook, and setting random seeds for consistent results.

20. **Q: What did you learn from this project?**

   - **A:** I gained experience in handling imbalanced datasets, optimizing model performance using evaluation metrics, and understanding the challenges of deploying machine learning models in real-world scenarios.

Project-Specific Questions

21. **Q: Why did you choose this project?**

   - **A:** I chose this project because it aligns with real-world challenges in data science, involves working with imbalanced datasets, and has significant societal and financial implications.

22. **Q: Can you explain your workflow for this project?**

   - **A:** The workflow included:

     - Data exploration and preprocessing.

     - Addressing data imbalance.

     - Building and evaluating machine learning models.

- Documenting results and conclusions.